A Shore Thing: Post-Hurricane Outcomes for Businesses in Coastal Areas

by

Melissa Chow
U.S. Census Bureau

Jordan Stanley
U.S. Census Bureau

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Abstract

During the twenty-first century, hurricanes, heavy storms, and flooding have affected many areas in the United States. Natural disasters and climate change can cause property damage and could have an impact on a variety of business outcomes. This paper builds upon existing research and literature that analyzes the impact of natural disasters on businesses. Specifically, we look at the differential effect of eight hurricanes during the period 2000-2009 on establishments in coastal counties relative to establishments in coastal-adjacent or inland counties. Our outcomes of interest include establishment employment and death. We find that following a hurricane event, establishments located in a coastal county have lower employment and increased probability of death relative to establishments in non-coastal counties.

Keyword*:* hurricane, natural disaster, coasts, employment, exit.

JEL Classification: D22, L20, Q54

[*] Chow is an economist at the U.S. Census Bureau, Center for Economic Studies. Stanley is an economist at the U.S. Census Bureau, Center for Enterprise Dissemination.

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I. INTRODUCTION

Natural disaster events have major ramifications for households and businesses in affected areas. In the twenty-first century, hurricanes and flooding have become increasingly alarming. In the United States, the Gulf of Mexico and Atlantic Coast have been particularly vulnerable to tropical storms. In August of 2017, Category 4 Hurricane Harvey caused about $125 billion (2017 dollars) in damages and was followed less than a month later by Hurricane Irma which caused damages estimated at $50 to $150 billion, according to the National Oceanic and Atmospheric Administration (NOAA).[1] Other major storm events that devastated parts of states and territories over the past twenty years include Hurricane Katrina (see Vigdor (2008) for a full discussion of the economic effects of Hurricane Katrina), Hurricane Sandy, and Hurricane Maria.

Combined with such storms, sea level rise due to climate change has greatly endangered coastlines around the world. Kulp and Strauss (2019) estimates that 250 million people currently live below projected annual flood levels with nearly 100 million more at risk based on mid-century projections. U.S. Census Bureau population estimates indicate that nearly 95 million Americans live in coastline regions with roughly 60 million people in areas most vulnerable to hurricanes, and the overall U.S. coastal population grew roughly 15% between 2000 and 2017.[2]

Given the concentration of people and businesses in coastal regions, it is important to assess the effects of hurricanes and flooding as well as how responsive individuals and

[1] https://census.gov/library/stories/2018/10/tracking-hurricanes-potential-impact-on-workforce-industries.html (Date accessed: November 15, 2018)
[2] https://www.census.gov/library/stories/2019/07/millions-of-americans-live-coastline-regions.html (Date accessed: October 8, 2019)

businesses are to actualized and perceived risks. Past research has examined the effect of hurricanes on individual economic outcomes such as employment and wage earnings (e.g. Belasen and Polachek (2009), Groen, Kutzbach, and Polivka (2020), and others). Other research has focused on broader ramifications of hurricanes on local economic activity at the county level (see Strobl (2011) and Boustan, Kahn, Rhode, and Yanguas (2020)) or in cities (e.g. Kocornik-Mina, McDermott, Michaels, and Rauch (2020)).

The present analysis is interested in the effects of hurricanes on businesses with a specific emphasis on differential effects by relative location. We extend elements of past work such as Strobl (2011), which examined economic growth in coastal areas following hurricanes, and Basker and Miranda (2018) which examined establishment-level outcomes in Mississippi following Hurricane Katrina. Our sample follows establishments from 2000 to 2009 and divides the entire coastal continental United States based on impact from major (Category 3 or higher) hurricanes in the given state and the location of the given county relative to the ocean. Specifically, we distinguish *coastal* counties as bordering the ocean, *coastal-adjacent* counties as bordering coastal counties but not touching the ocean, and *inland* counties as bordering neither coastal counties nor the ocean.

We are particularly interested in differential effects by relative location in response to both actualized risk (i.e. the hurricane striking an area) and perceived risk (i.e. coasts then being seen as riskier in the long run due to storms, sea level rise, and flooding). Regarding the latter, we seek to assess whether hurricane events have effects beyond a mechanical reaction in the immediate aftermath of the storm. For example, Filippova, Nguyen, Noy, and Rehm (2020) examines whether information disclosure about sea level rise and the future risk of coastal living affects property prices. That study found insignificant effects, suggesting that people did not

factor in long-term risks as house prices showed no risk premium. Thinking about businesses, if and how establishments show signs of recovery in the years following a hurricane could signal how risk is perceived. Past work such as Messner and Meyer (2006) cites the importance of considering risk perception when performing flood damage research.

In our study, we analyze differential responses by relative location in addition to post-hurricane effects. Coastal areas could be seen as riskier along several dimensions (sea level rise, storm strength, flooding, etc.); comparing estimated effects across locations relative to the ocean can help assess direct effects of hurricanes as well as resulting changes in perceived economic viability on the coast. Our empirical strategy provides estimates of economic effects for establishments in coastal counties hit by a hurricane relative to both establishments located more inland in the same state as well as establishments in coastal counties in states that were not hit by a hurricane during our sample period.

We use the U.S. Census Bureau's Longitudinal Business Database (LBD) as our primary data source. The hurricane information is taken from NOAA and Strobl (2011). Our county-level information is derived from data from the U.S. Census Bureau. Methodologically, we employ a triple differences strategy exploiting variation in location relative to the ocean, presence in a hurricane-hit state, and time relative to hurricane event. An intuitively similar strategy was used in Henninghausen and Suter (2020) to analyze the effect of flooding in Colorado on housing prices. The establishment outcomes we analyze are employment and death.

Our findings indicate that establishments in coastal counties in states hit by a hurricane have lower employment in the post-hurricane period compared to similar establishments in coastal-adjacent or inland counties. Further, the probability of establishment death post-hurricane is greater for such establishments. For establishments in coastal-adjacent counties, the

relative effects for employment are statistically insignificant, while the probability of death post-hurricane is higher for such establishments. Comparing the effects in coastal and coastal-adjacent counties implies the size and statistical significance of effects of hurricanes increase for establishments closer to the coast.

The paper proceeds with more thorough descriptions of the data (Section 2) and methodology (Section 3). Then, we present our regression results (Section 4) and discuss implications and conclusions (Section 5).

II. DATA

Our source for information on establishment and firm characteristics is the Census Bureau's Longitudinal Business Database (LBD). The LBD is the longitudinal version of the Business Register that has records for all private, non-farm employers, in all industries and geographically covers all U.S. states (Jarmin and Miranda, 2002; Stinson, White, and Lawrence, 2017). It includes data on establishment employment, births, deaths, establishment age, firm age, firm size, industry codes, and establishment geography details. The data are available annually at the establishment level from 1976 through 2016.

The two outcome variables used in our regressions are employment and an indicator for establishment death. Annual employment data are collected on March 12th for the year that they are reported. The death variable is equal to 1 if the year is the last year that the establishment shows up in the LBD within our time period and 0 otherwise. Control variables used in our regressions include the following variables: establishment age, firm age, firm size, industry codes, and geographic variables, which provide information on business characteristics and geography. Establishment age is the number of years that the establishment has been in

business. For firms that are single units, firm age is equal to the establishment age. For firms that are comprised of multiple units, firm age is defined as the age of the oldest establishment. Firm size is measured as the number of establishments that make up the firm. Even though our time period of analysis is after the SIC-NAICS transition, NAICS industry codes still undergo some changes every five years. Therefore, to maintain consistent use of a single vintage of NAICS industry codes, we use the vintage consistent NAICS codes (Fort and Klimek, 2016). Geographic variables, such as county and state, are used to merge the LBD with the datasets listed below, and the state variable is used to determine whether an establishment is located in a hurricane-hit state.

Hurricane data come from the National Oceanic and Atmospheric Administration (NOAA). The hurricanes included in our analysis sample are those categorized as Category 3 or higher in the time period from 2000 through 2009.[3] Additionally, we enhance the NOAA list with the list of hurricanes and states affected from Strobl (2011). Strobl (2011) uses new wind speed data and HAZUS software from FEMA to model the tracks of hurricanes, to calculate maximum wind speed, and to determine the states affected in the coastal county sample. A list of the hurricanes and states affected in our sample are displayed in Table 1. During the time period of our analysis, all Category 3+ hurricanes occurred in the years 2004 and 2005.

[3] Historical information on hurricanes is available here: https://www.aoml.noaa.gov/hrd/hurdat/All_U.S._Hurricanes.html (Date accessed: December 17, 2018)

Table 1: Hurricanes and States Affected

Hurricane Name	Year	States Affected
Jeanne	2004	FL, GA
Ivan	2004	AL, FL, LA, MS
Frances	2004	FL, GA
Charley	2004	FL
Wilma	2005	FL, SC, VA
Rita	2005	LA, MS, TX
Katrina	2005	AL, FL, LA, MS
Dennis	2005	AL, FL

Notes: This table lists the Category 3+ hurricanes that occurred during our analysis period (2000 – 2009) and the states that were affected by the hurricanes. It combines data from NOAA with supplemental information from Strobl (2011).

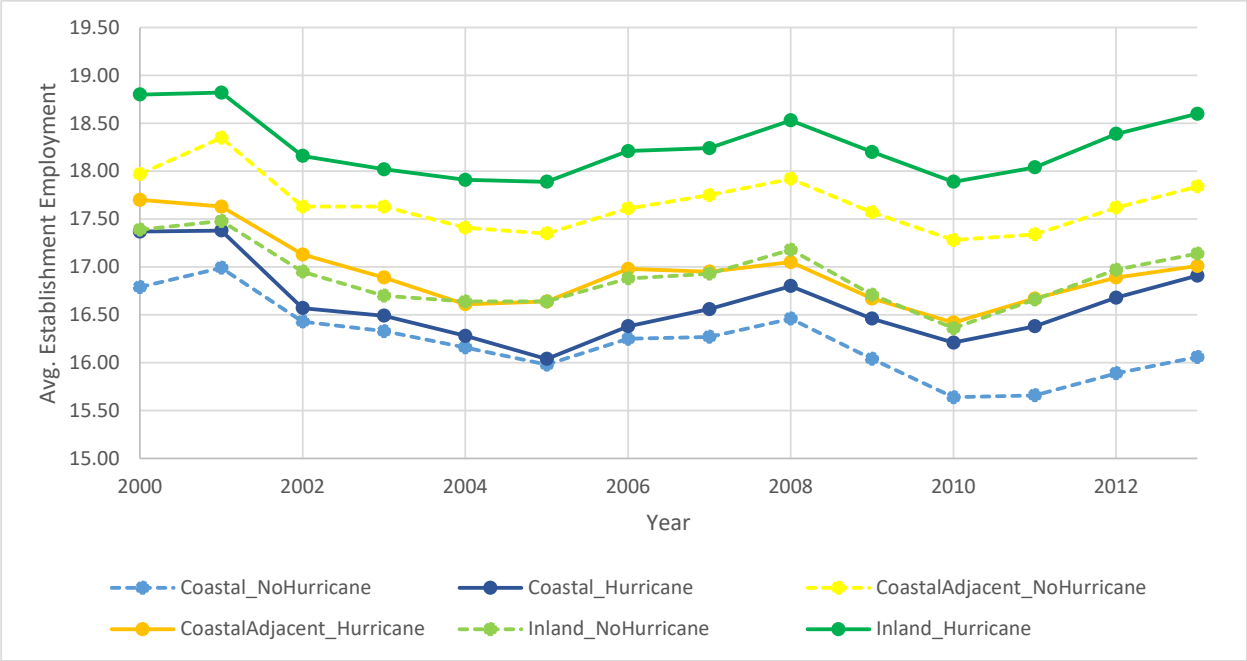
Our coastal county definitions are compiled based on information from the U.S. Census Bureau and county delineations from 2015.[4] County population data are from the U.S. Census Bureau compiled by the National Bureau of Economic Research (NBER).[5] This dataset contains annual population data for each county in the U.S. from 1970 – 2014. These data are merged with the LBD using a county identifier. Once these datasets are merged, indicator variables and interaction terms are generated for each observation. Each county is classified into one of three groups: Coastal, Coastal-Adjacent, and Inland. Coastal counties are defined as those that touch the ocean. Coastal-adjacent counties are counties that neighbor coastal counties, but do not touch the ocean. Inland counties touch neither the ocean nor coastal counties. Each state is designated as either "Hurricane" or "Non-Hurricane" depending on whether or not a Category 3+ hurricane hit the state during the analysis time period. There are also variables to indicate the

[4] Information and maps with county designations are found on: https://www.census.gov/geo/maps-data/maps/county_wall.html (Date accessed: October 9, 2018)
[5] For more information about the county population data, see https://data.nber.org/data/census-intercensal-county-population.html (Date accessed: March 9, 2019)

time period relative to when a hurricane hits. Additional details about the post variable are included in Section 3. Interaction terms are constructed based on the coastal indicator variable, the hurricane indicator variable, and the post-hurricane indicator variable.

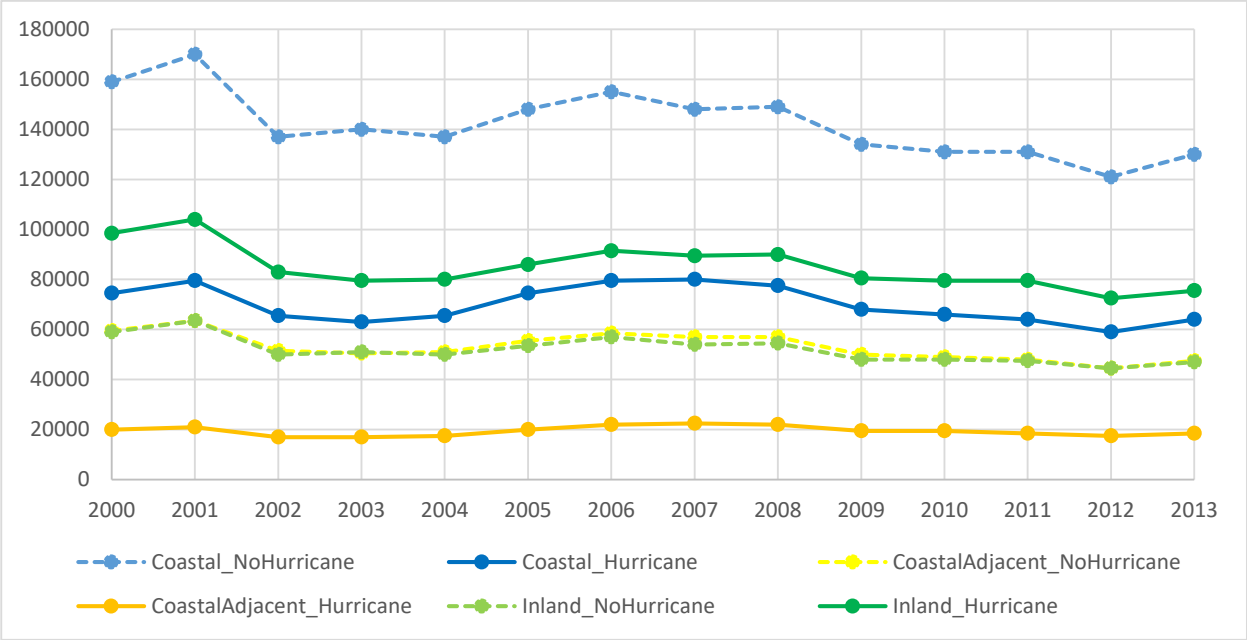
Trends over time (from 2000 through 2013) for average establishment employment and total establishment deaths are displayed in Figures 1 and 2. For each geographic group, the solid line represents the states that were hit by hurricanes while the dashed line represents the states that were not hit by hurricanes. Lines of the same color represent the three different coastal geographic groups. The blue lines indicate establishments in the coastal group, the orange/yellow lines show the coastal-adjacent group, and the green lines represent the inland group. Figure 1 shows average annual employment for each of the six geographic groups (geographic group × hurricane status). While all groups show parallel trends in the time period prior to the hurricane, employment trends for the coastal groups converge during the hurricane time period (see Figure 1). In the years following the major hurricane events (2006 onward), the employment trends for the coastal groups diverge following the hurricane with the coastal-hurricane group increasing employment at a higher level and rate than the coastal-non-hurricane group (see Figure 1). Figure 2 shows total establishment deaths over time within each one of our geographic categories. Compared to such trends in areas that do not border the ocean, trends in establishment deaths throughout the time period for the coastal groups fluctuate more (see Figure 2).

Figure 1: Average Establishment Employment by Geographic Group and Hurricane State



Notes: This figure presents average establishment employment from 2000 through 2013 for each county geography and hurricane state combination. "Coastal" is a binary variable that is "1" if the establishment is located in a county that borders the ocean and 0 otherwise. "Coastal-Adjacent" represents a binary variable that is "1" if the establishment is in a county that does not border an ocean but does border an ocean-bordering county. "Inland" is a binary variable that is 1 if the establishment is located in a non-Coastal and non-Coastal-Adjacent county. "Hurricane" represents a binary variable that is "1" if the establishment is located in a state that was hit by a Category 3 or higher hurricane in 2004 or 2005. Disclosure Approval Number: CBDRB-FY20-CES007-005.

Figure 2: Total Deaths by Geographic Group and Hurricane State



Notes: This figure presents total establishment deaths from 2000 through 2013 for each county geography and hurricane state combination. "Coastal" is a binary variable that is "1" if the establishment is located in a county that borders the ocean and 0 otherwise. "Coastal-Adjacent" represents a binary variable that is "1" if the establishment is in a county that does not border an ocean but does border an ocean-bordering county. "Inland" is a binary variable that is 1 if the establishment is located in a non-Coastal and non-Coastal-Adjacent county. "Hurricane" represents a binary variable that is "1" if the establishment is located in a state that was hit by a Category 3 or higher hurricane in 2004 or 2005. Disclosure Approval Number: CBDRB-FY20-CES007-005.

The trends depicted in Figures 1 and 2 could reflect other factors that differ across geographic groups. As such, Table 2 presents establishment-level summary statistics by group for the following control variables: establishment age, age of firm, and the number of establishments in the firm to which the establishment belongs (i.e. firm size). The means for firm age and number of establishments are establishment-weighted with an average firm age of 13 years and an average number of establishments per firm of 250. On average, establishments in our sample are roughly 10 years old. Across all geographic groups, establishments in states not hit by a hurricane are slightly older than establishments in states that are hit by a hurricane.

Establishments in states that are hit by a hurricane are part of firms that are larger on-average than establishments in the non-hurricane states.

Table 2: Establishment-Level Summary Statistics, 2003

	Full Sample	Inland - No Hurricane	Inland - Hurricane	Coastal Adjacent - No Hurricane	Coastal Adjacent - Hurricane	Coastal - No Hurricane	Coastal - Hurricane
Establishment Age	10.29	11.01	10.21	10.61	9.39	10.38	9.47
Firm Age	13.14	13.88	13.64	13.38	12.78	12.73	12.62
Establishments in Firm	253	250	292	243	282	223	278
N	3,863,000	522,000	793,000	510,000	163,000	134,000	535,000

Notes: This table presents establishment means in 2003 for the following variables: establishment age, firm age, and the number of establishments in the firm to which the establishment belongs. The means for firm age and establishments in firm are establishment-weighted. It also includes the number of observations for each geography by hurricane state group. "Coastal" is a binary variable that is "1" if the establishment is located in a county that borders the ocean and 0 otherwise. "Coastal-Adjacent" represents a binary variable that is "1" if the establishment is in a county that does not border an ocean but does border an ocean-bordering county. "Inland" is a binary variable that is 1 if the establishment is located in a non-Coastal and non-Coastal-Adjacent county. "Hurricane" represents a binary variable that is "1" if the establishment is located in a state that was hit by a Category 3 or higher hurricane in 2004 or 2005. Establishment age and firm age are measured in number of years and firm size is measured as the number of establishments. Disclosure Approval Number: CBDRB-FY20-CES007-005.

Table 3 displays the sector breakdown and the top five sectors represented in each of the groups. The top five sectors are the same for all the groups: Retail; Professional, Scientific, and Technical Services; Health Care and Social Assistance; Other Services (except Public Administration); and Construction. However, the top five sectors are in different orders depending on the group. The largest sector represented in the coastal-adjacent and inland groups, is the retail sector. For the coastal groups, the professional services sector is the largest, with the retail sector as the second largest.

Table 3: Industrial Sector Breakdown, 2003

	Full Sample	Inland - No Hurricane	Inland - Hurricane	Coastal Adjacent - No Hurricane	Coastal Adjacent - Hurricane	Coastal - No Hurricane	Coastal - Hurricane
Largest Industry	Retail	Retail	Retail	Retail	Retail	Professional	Professional
2nd	Professional	Construction	Other Services	Professional	Professional	Retail	Retail
3rd	Healthcare	Healthcare	Professional	Healthcare	Other Services	Healthcare	Healthcare
4th	Other Services	Other Services	Healthcare	Construction	Construction	Other Services	Other Services
5th	Construction	Professional	Construction	Other Services	Healthcare	Construction	Construction

Notes: This table presents the industrial sector breakdown in 2003 by listing the top 5 largest industries (by number of establishments) for the full sample and each of the county geography and hurricane state combinations. "Coastal" is a binary variable that is "1" if the establishment is located in a county that borders the ocean and 0 otherwise. "Coastal-Adjacent" represents a binary variable that is "1" if the establishment is in a county that does not border an ocean but does border an ocean-bordering county. "Inland" is a binary variable that is 1 if the establishment is located in a non-Coastal and non-Coastal-Adjacent county. "Hurricane" represents a binary variable that is "1" if the establishment is located in a state that was hit by a Category 3 or higher hurricane in 2004 or 2005. Disclosure Approval Number: CBDRB-FY20-CES007-005.

III. METHODOLOGY

Our empirical strategy is a triple difference methodology. The independent variables of interest are the coastal county indicator, the indicator for presence in a state hit by a Category 3 or higher hurricane in 2004 or 2005, and the pre/post hurricane indicator. The main effect of interest is the coefficient estimate for the triple interaction term. The outcomes analyzed are the natural logarithm of establishment employment and an indicator for establishment "death". A linear probability model is used for the death regressions. The general regression model is presented below in Equation 1:

$$[1] \quad Outcome_{icst} = \alpha_{icst} + \beta_1 Coastal_c \times Post_t \times Hurr_s + \beta_2 Coastal_c \times Post_t + \beta_3 Coastal_c \times Hurr_s$$

$$+ \beta_4 Post_t \times Hurr_s + \beta_5 Coastal_c + \beta_6 Post_t + \beta_7 Hurr_s + \psi X_i + \rho Pop_{ct} + \varphi_s + \tau_t + \varepsilon_{icst}$$

Each outcome is an establishment-year observation. The "death" outcome indicates that an establishment exited the data set and never returned. Note that the "death" model is thus a

hazard model (no observations after death). Establishment employment is logged; therefore, establishments with zero employment in year *t* are not in that regression sample. However, such establishments could be present in the death regression sample.

The triple interaction term indicates the post-hurricane effect of being located in a coastal county in a state hit by a hurricane. As this is the primary effect we seek to estimate, β_1 is our main coefficient of interest. "Coastal" is equal to 1 if the establishment is in a county that borders the ocean and 0 otherwise. So for this specification, the comparison group is establishments in coastal-adjacent or inland counties as defined in Section 2. "Hurr" is a state-level variable indicating whether or not a Category 3 or higher hurricane hit the state during our sample period.

"Post" is a time-based binary indicator for year *t* being before or after the year of the relevant hurricane event. For establishments in states that were not hit by a major hurricane during our sample period, the "post" period is 2006 onward. For establishments in states hit by a major hurricane, the "post" period varies depending on when the earliest major hurricane occurred and what the dependent variable is. To summarize succinctly here: the "post" period starts in the year after the earliest hurricane for the employment regressions (e.g. hurricane in 2005 means "Post" is 1 for 2006 onward); for the death regressions, the "post" period begins in the year the hurricane hit (e.g. hurricane in 2005 means "Post" is 1 for 2005 onward).

All regression specifications include variables for county population ("Pop_{ct}"), state-level fixed effects (φ_s), and calendar year fixed effects (τ_t). The time-variant control variables included in **X** are establishment age, establishment industry, firm age, and firm size. Establishment industry is the two-digit sector, and firm size is number of establishments for the firm identification code. In presenting our results, we show the estimates from specifications

both including and excluding the business characteristic variables included in **X**. Put another way, the baseline results only have the geographic and time controls in addition to the coastal-related variables of interest. Our preferred specifications are those which include all the business characteristic variables.

We run a second specification where we split out an indicator variable and the accompanying interaction terms for coastal-adjacent county along with the coastal county variables. This model is shown in Equation 2:

$$[2] \quad Outcome_{icst} = \alpha_{icst} + \beta_1 Coastal_c \times Post_t \times Hurr_s + \beta_2 Coastal_c \times Post_t + \beta_3 Coastal_c \times Hurr_s + \beta_4 Coastal_c + \beta_5 CoastalAdj_c \times Post_t \times Hurr_s + \beta_6 CoastalAdj_c \times Post_t + \beta_7 CoastalAdj_c \times Hurr_s + \beta_8 CoastalAdj_c + \beta_9 Post_t \times Hurr_s + \beta_{10} Post_t + \beta_{11} Hurr_s + \psi X_i + \rho Pop_{ct} + \varphi_s + \tau_t + \varepsilon_{icst}$$

Here, the comparison group consists solely of establishments in inland counties. This allows us to compare effects more directly for establishments in coastal versus coastal-adjacent counties. Effectively, the coefficient estimates from Equation 1 show the average effect of being an establishment in a county that borders the ocean compared a similar establishment in a county that does not border the ocean. The coefficient estimates from Equation 2 would thus show average effects for establishments in counties that border the ocean compared to similar establishments in inland counties (at least two counties away from the ocean) as well as average effects for establishments in counties bordering counties that border the ocean compared to similar establishments in farther inland counties.

IV. REGRESSION RESULTS

The regression sample runs from the year 2000 through 2009 and includes establishments in all coastal states in the continental United States. Standard errors are clustered at the county level. The main findings of interest are presented in Tables 4 and 5. As the primary focus of our analysis is on differential effects for establishments in coastal counties, we include the coefficient estimates for all coastal-related variables. A separate table was created for each outcome variable. The table columns are presented to show the estimates from each of our specifications of interest. Columns (1) and (2) show the coefficient estimates from the coastal vs. non-coastal regressions (Equation 1), while Columns (3) and (4) show the coefficient estimates from the specifications that contain separate independent variables for coastal and coastal-adjacent (Equation 2). Columns (1) and (3) are the results from the baseline specifications, and Columns (2) and (4) are results from the full specifications which include business characteristics. Our preferred specification corresponds to the results in Column (4), so our discussion will focus on those estimates with estimates from the other specifications providing useful reference points.

The employment effects are presented in Table 4. Across specifications, the results show negative and statistically significant effects on post-hurricane employment for establishments in coastal counties in hurricane-hit states across specifications (see row for "Coastal×Hurr×Post"). The effects are approximately 2 log points, or roughly 2 percentage point lower employment. Most of these estimated effects are statistically insignificant for establishments in coastal-adjacent counties (see row for "CoastalAdj×Hurr×Post"). The estimated effects for the "Coastal×Post", "Coastal×Hurr", and "Coastal" variables as well as the corresponding "Coastal-Adjacent" variables are statistically insignificant at the 5% level when including all controls (see

Columns (2) and (4)). In our preferred specification (Column (4)), there is a weakly significant effect of "Coastal×Post", implying a roughly 0.63 percentage point drop in employment in coastal counties in the post-hurricane period across all states.

Table 4: Employment Regression Results

	(1)	(2)	(3)	(4)
Coastal×Hurr×Post	-0.0258***	-0.0212***	-0.0264**	-0.0185***
	(0.0099)	(0.0061)	(0.0105)	(0.0062)
Coastal×Post	-0.0108*	-0.0045	-0.0125*	-0.0063*
	(0.0061)	(0.0034)	(0.0064)	(0.0033)
Coastal×Hurr	0.0882***	0.0384*	0.1113***	0.0375
	(0.0281)	(0.0225)	(0.0341)	(0.0267)
Coastal	-0.0790***	-0.0188	-0.0879***	-0.0001
	(0.0187)	(0.0151)	(0.0297)	(0.0204)
CoastalAdj×Hurr×Post			-0.0198*	-0.0053
			(0.0119)	(0.0078)
CoastalAdj×Post			-0.0033	-0.0037
			(0.0048)	(0.0042)
CoastalAdj×Hurr			0.0663	0.0251
			(0.0428)	(0.0283)
CoastalAdj			-0.0159	0.0333*
			(0.0254)	(0.0172)
Establishment/Firm Controls	No	Yes	No	Yes

Notes: This table presents select coefficient estimates for the effects of relative location on the natural logarithm of employment with standard errors clustered at the county level. The regression models used are shown in Equation 1 and Equation 2 in Section 3 of the paper. "Coastal" is a binary variable that is "1" if the establishment is located in a county that borders the ocean and 0 otherwise. "CoastalAdj" represents a binary variable that is "1" if the establishment is in a county that does not border an ocean but does border an ocean-bordering county. "Hurr" represents a binary variable that is "1" if the establishment is located in a state that was hit by a Category 3 or higher hurricane in 2004 or 2005. "Post" is a binary variable that equals "1" if year *t* is after the year of the hurricane event. Establishment/Firm Controls include establishment age, establishment industry (2-digit sector), firm age, and firm size (total number of establishments). Disclosure Approval Number: CBDRB-FY20-CES007-005.

Table 5 contains the estimated effects on establishment death. In our preferred specification (see Column (4)), the probability of an establishment in our sample dying post-hurricane is roughly 0.7 percentage points higher for establishments in coastal counties and 0.44 percentage points higher for establishments in coastal-adjacent counties. For the other coastal-

related variables, establishments in coastal and coastal-adjacent counties show roughly 0.4 and 0.35 percentage point higher probabilities of death relative to establishments in inland counties (see Column (4) for rows for "Coastal" and "CoastalAdj"). The "Coastal×Hurr" and "CoastalAdj×Hurr" interaction terms show roughly 0.58 and 0.74 percentage point lower probabilities of death for establishments in such counties. This implies that in the pre-hurricane period, establishments in coastal and coastal-adjacent counties in states eventually hit by a major hurricane were less likely to die compared to corresponding establishments in such areas in hurricanes that were not hit by a major hurricane in our sample period.

Table 5: Death Regression Results

	(1)	(2)	(3)	(4)
Coastal×Hurr×Post	0.0070***	0.0065***	0.0078***	0.0070***
	(0.0025)	(0.0023)	(0.0025)	(0.0023)
Coastal×Post	-0.0020	-0.0022*	-0.0017	-0.0020
	(0.0013)	(0.0012)	(0.0013)	(0.0012)
Coastal×Hurr	-0.0056***	-0.0031*	-0.0094***	-0.0058***
	(0.0021)	(0.0018)	(0.0024)	(0.0021)
Coastal	0.0047***	0.0019	0.0082***	0.0039**
	(0.0015)	(0.0013)	(0.0017)	(0.0016)
CoastalAdj×Hurr×Post			0.0057***	0.0044**
			(0.0023)	(0.0021)
CoastalAdj×Post			0.0007	0.0005
			(0.0016)	(0.0015)
CoastalAdj×Hurr			-0.0089***	-0.0074***
			(0.0024)	(0.0021)
CoastalAdj			0.0062***	0.0034**
			(0.0016)	(0.0014)
Establishment/Firm Controls	No	Yes	No	Yes

Notes: This table presents select coefficient estimates for the effects of relative location on the probability of an establishment in our sample dying with standard errors clustered at the county level. The regression models used are shown in Equation 1 and Equation 2 in Section 3 of the paper. "Coastal" is a binary variable that is "1" if the establishment is located in a county that borders the ocean and 0 otherwise. "CoastalAdj" represents a binary variable that is "1" if the establishment is in a county that does not border an ocean but does border an ocean-bordering county. "Hurr" represents a binary variable that is "1" if the establishment is located in a state that was hit

by a Category 3 or higher hurricane in 2004 or 2005. "Post" is a binary variable that equals "1" if year *t* is after or equal to the year of the hurricane event. Establishment/Firm Controls include establishment age, establishment industry (2-digit sector), firm age, and firm size (total number of establishments). U.S. Census Bureau Disclosure Approval Number: CBDRB-FY20-CES007-005.

V. DISCUSSION

Overall, our results imply statistically significant differential economic outcomes for establishments in coastal counties compared to similar establishments in coastal-adjacent and inland counties. Lower employment post-hurricane aligns with past findings, such as Belasen and Polachek (2009). In our analysis, effects on employment are only statistically significant for establishments in coastal counties. The employment effects for establishments in non-coastal counties (coastal-adjacent or inland counties) are insignificant. Relative to establishments in inland counties, several of the significant effects for establishments in coastal counties are likewise significant for establishments in coastal-adjacent counties. Strobl (2011) does not distinguish between coastal and coastal-adjacent counties; however, the findings of lower economic growth rates in coastal counties after a hurricane intuitively fit with the estimated effects from our analysis.

Based on our present findings, the estimated economic effects of hurricanes do seem to be stronger for establishments in geographic areas bordering the ocean compared to those in areas further inland. Differential employment effects are only statistically significant for establishments in coastal counties, while business dynamics are significantly affected in both coastal and coastal-adjacent counties relative to inland counties. One possible implication of this is that actualized risks are more relevant to establishments closest to the coastline while perceived risks have broader effects. Establishments in coastal-adjacent counties exhibit no significant employment effects in our preferred specification but have a higher probability death.

Additionally, coastal counties have a higher probability of death compared to establishments in coastal-adjacent counties.

A few mechanisms could be relevant to these estimated effects. First, hurricanes may simply have a larger impact on establishments on the coast compared to those more inland – the strength of the effects increases moving from inland counties to coastal-adjacent counties to coastal counties. Another potential explanation is that the direct economic effects of hurricanes are mostly felt among establishments in coastal counties with some spillover effects into counties further inland. Finally, perceived risks following these hurricane events could vary by location relative to the ocean. For example, if being on the coast is seen as riskier, surviving establishments may have trouble retaining workers or bringing in new hires.

VI. CONCLUSION

As hurricanes and rising sea levels become more prevalent events, it becomes increasingly important to understand what impact, if any, these events have on businesses. Results from the current analysis suggest that establishments in coastal counties are on-average more significantly affected by hurricanes relative to counties further inland. The estimated effects on establishment employment and deaths in coastal counties are all statistically significant. Relative to non-coastal counties, establishments located in coastal counties in hurricane-hit states have decreased employment and higher probability of death following a hurricane. Relative to further inland counties, establishments in coastal-adjacent counties (those bordering counties that border the ocean) also exhibit a higher probability of death in the post-hurricane years.

We expect to extend the present analysis by performing several steps to further refine our assessment of the impact of business location in coastal counties on outcomes such as employment and establishment death. We also hope to perform supplemental analysis on establishment births. Since our current dataset only includes establishments that already exist, it is difficult to tease out the probability of establishment birth. Instead, to look at births, we could perform an analysis at the county level using number of births as our outcome variable. Another alternative analysis could be done using a dataset of business applications to estimate the probability of birth.

In extending this research, we plan to run an event study analysis, complete several robustness checks, and attempt to improve establishment links in the microdata to determine whether the births and deaths are true births and deaths. The current analysis only separates the analysis time period into pre- and post-hurricane time periods. An event study framework would allow us to look at trends each year before the hurricane occurs and then allow an analysis of each year following the event. We could then examine whether the effects are sustained over time or if they decrease with each incremental year following a hurricane hitting a state.

Some robustness checks of our analysis include using NOAA's definition of coastal counties and its strict definition of hurricanes that are Category 3 or higher. NOAA does not base its definition of a coastal county on relative location to the ocean. Instead, coastal counties are defined based on whether a county meets one of the following criteria: "(1) at least 15 percent of a county's total land area is located within the Nation's coastal watershed; or 2) a portion of or an entire county accounts for at least 15 percent of a coastal cataloging unit."[6] The

[6] NOAA definitions for coastal counties are explained more in-depth in the following document: https://www.census.gov/geo/landview/lv6help/coastal_cty.pdf (Date accessed: October 29, 2018)

NOAA definition of coastal counties also includes counties that border the Great Lakes. For the purpose of our analyses, we only consider coastal counties as those that touch an ocean. We could check whether using NOAA's coastal counties definition for our sample of states affects our results. An additional robustness check would be to only use NOAA's list of Category 3+ hurricanes. Currently, we supplement the list of NOAA Category 3+ hurricanes with the hurricane list from Strobl (2011).

Further steps in our analysis include expanding our data with the revenue measures included the new version of the LBD. This dataset will allow us to include firm revenue measures as additional variables for our regressions. Additionally, there has been work to improve establishment linkages in the microdata. Using these new linkages could help us understand whether establishments are truly new ones or whether they are continuers that have been inactive for a few years and then return to being active. Similarly, improved linkages could help us better identify deaths.

REFERENCES

Basker, E. and Miranda, J., 2018. Taken by storm: business financing and survival in the aftermath of Hurricane Katrina. *Journal of Economic Geography*, *18*(6), pp.1285-1313.

Belasen, A.R. and Polachek, S.W., 2009. How disasters affect local labor markets the effects of hurricanes in Florida. *Journal of Human Resources*, *44*(1), pp.251-276.

Boustan, L.P., Kahn, M.E., Rhode, P.W. and Yanguas, M.L., 2020. The effect of natural disasters on economic activity in US counties: A century of data. *Journal of Urban Economics*, Vol. 118, Article 103257.

Cohen, D. "About 60.2M Live in Areas Most Vulnerable to Hurricanes". *America Counts: Stories Behind the Numbers.* Published online July 15, 2019. https://www.census.gov/library/stories/2019/07/millions-of-americans-live-coastline-regions.html

Dowell, E. "Census OnTheMap Tool Helps Measure Impact of Disasters". *America Counts: Stories Behind the Numbers.* Published online October 5, 2018. https://census.gov/library/stories/2018/10/tracking-hurricanes-potential-impact-on-workforce-industries.html

Filippova, O., Nguyen, C., Noy, I. and Rehm, M., 2020. Who Cares? Future Sea Level Rise and House Prices. *Land Economics*, *96*(2), pp.207-224.

Fort, T., and Klimek, S., 2016. The Effects of Industry Classification Changes on US Employment Composition. Unpublished paper, U.S. Census Bureau.

Groen, J.A., Kutzbach, M.J. and Polivka, A.E., Storms and Jobs: The Effect of Hurricanes on Individuals' Employment and Earnings over the Long Term. *Journal of Labor Economics.* Forthcoming.

Henninghausen, H., and Suter, J.F., 2020. Flood Risk Perception in the Housing Market and the Impact of a Major Flood Event. *Land Economics, 96*(3), pp. 366-383.

Jarmin, R. S., and Miranda, J., 2002. The Longitudinal Business Database. Unpublished paper, U.S. Census Bureau.

Kocornik-Mina, A., McDermott, T.K., Michaels, G. and Rauch, F., 2020. Flooded cities. *American Economic Journal: Applied Economics*, *12*(2), pp.35-66.

Kulp, S.A., Strauss, B.H., 2019. New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding. *Nature Communications* 10, 4844.

Messner, F. and Meyer, V., 2006. Flood damage, vulnerability and risk perception–challenges for flood damage research. In *Flood risk management: hazards, vulnerability and mitigation measures* (pp. 149-167). Springer, Dordrech

Stinson, M., White, T.K., and Lawrence, J., 2017. Upcoming Improvements to the Longitudinal Business Database and Business Dynamics Statistics. Unpublished paper, U.S. Census Bureau.

Strobl, E., 2011. The economic growth impact of hurricanes: Evidence from US coastal counties. *Review of Economics and Statistics*, *93*(2), pp.575-589.

Vigdor, J., 2008. The economic aftermath of Hurricane Katrina. *Journal of Economic Perspectives*, *22*(4), pp.135-54.